EdgeSign: An IoT-Enabled Edge–Cloud Hybrid System for Real-Time Sign Language Translation

MAJOR PROJECT VALIDATION FORM

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1	Abstract	Real-time gesture-to-speech systems play a crucial role in improving communication accessibility for individuals with speech and hearing impairments. The proposed assistive communication framework is implemented using Raspberry Pi 4 along with the Raspberry Pi Camera Module 3 to capture continuous hand gesture movements. MediaPipe Hands is utilized to extract twenty-one key hand landmarks, which are further processed to classify gestures with high accuracy. Unlike conventional approaches that translate gestures into isolated words, the system integrates an AI-driven sentence-generation mechanism to construct meaningful and grammatically coherent sentences from gesture sequences. Text-to-speech synthesis is performed on the Raspberry Pi to produce audible communication output. The system achieves low latency while maintaining portability and cost efficiency, making it suitable for educational, clinical, and social environments. Experimental evaluation confirms the reliability of the gesture recognition pipeline and demonstrates the system's capability to function as a standalone assistive communication solution.

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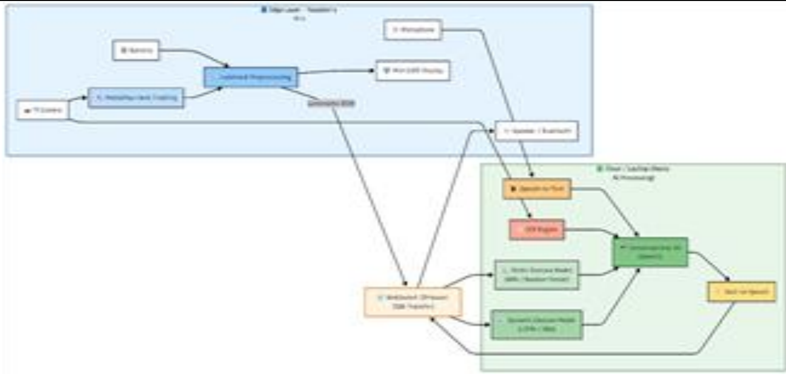
2	RESEARCH GAP	1. Existing systems depend heavily on cloud processing, leading to latency and reduced real-time performance on low-cost edge devices. 2. Most sign language recognition systems provide only word-level output without contextual or sentence-level understanding. 3. Available models are trained on fixed datasets and lack mechanisms for on-device data collection and retraining in real-world environments.
3	Please furnish the details of 5 research articles in IEEE Reference format with Impact factor and year of publication from Which Research gap has been identified (F. Lastname, “Title of article,” *Journal Name*, vol., no., pp., year. And IF 3.1)) / 5 Patent Details for highlighting the innovation	1. Google Research, "MediaPipe Hands: Real-time hand tracking for mobile and edge devices," Google AI Research, 2020. 2. Raspberry Pi Foundation, "Raspberry Pi 4 Model B technical specifications and architecture," Raspberry Pi Documentation, 2019. 3. S. Mitra and T. Acharya, "Gesture recognition: A survey," IEEE Transactions on Systems, Man, and Cybernetics, vol. 37, no. 3, pp. 311–324, 2007. 4. Y. Zhang, F. Chen, and L. Wang, "Vision-based hand gesture recognition using deep learning techniques," IEEE Access, vol. 8, pp. 112345–112360, 2020. 5.] A. Graves, A. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 6645–6649, 2013.
4	Bridging the Research Gap/ Innovation Gap	1. Introduces an edge–cloud hybrid architecture to reduce latency and computation load on embedded devices. 2. Enables real-time gesture-to-speech translation with sentence generation, unlike isolated word-based systems. 3. Supports low-cost, portable deployment using Raspberry Pi for practical assistive communication.
5	OBJECTIVE	1. To design a real-time sign language recognition system using vision-based hand tracking. 2. To implement an edge-based gesture classification model optimized for Raspberry Pi. 3. To generate meaningful text and speech output from recognized gestures. 4. To integrate edge–cloud collaboration for improved accuracy and contextual understanding. 5. To develop a portable, low-power assistive communication device for real-world use.
6	System Architecture	

		
7	METHODOLOGY	1. Capture hand gestures using Raspberry Pi Camera Module. 2. Extract 21 hand landmarks using MediaPipe Hands. 3. Preprocess landmark data into structured numerical vectors. 4. Store gesture data locally for training and testing datasets. 5. Train a Random Forest classifier for gesture recognition. 6. Perform real-time gesture inference on the Raspberry Pi (edge). 7. Transmit recognized gesture data to cloud using WebSocket. 8. Generate contextual sentences using cloud-based AI (Gemini). 9. Convert generated text to speech using TTS module. 10. Display output via speaker, mobile app, or OLED display.
8	OUTCOMES OR DELIVERABLES	1. Real-time sign language gesture recognition system. 2. Accurate gesture-to-text and gesture-to-speech conversion. 3. Edge–cloud hybrid architecture for low-latency processing. 4. Mobile app interface for customization and monitoring. 5. Fully functional prototype with documentation and results.
9	NOVELTY	1. Edge–cloud hybrid processing using lightweight landmark data. 2. Random Forest–based gesture recognition optimized for edge devices. 3. Context-aware sentence generation using cloud AI integration.
10	Time line	Deliverables
	November 2025(Review 0)	Problem definition, literature survey, requirement analysis
	January 2026 (SPRINT Review 1)	Hardware setup, MediaPipe integration, dataset creation

	March 2026 (SPRINT Review 2)	Gesture recognition model training, cloud integration
	April 2026 (SPRINT Review 3)	Testing, optimization, final demo & documentation
11	Collaboration details	1.Surya Sathyanarayanan 2.Jeevan Baabu Murugan
12	Target **Technology Readiness Level of the project** (TRL Scale 1 to 9) TRL 1 -Basic principles observed and reported. TRL 2 -Technology concept formulated. TRL 3 -Experimental proof of concept. TRL 4- Technology validated in lab. TRL 5- Technology validated in a relevant environment. TRL 6 -Technology demonstrated in a relevant environment. TRL 7- System prototype demonstration in operational environment. TRL 8- System complete and qualified. TRL 9- Actual system proven in operational environment.	Target TRL: TRL 6 – Technology demonstrated in a relevant environment • The core system functionalities such as hand gesture capture, landmark extraction, gesture classification using a Random Forest model, and edge–cloud communication have been implemented and tested. • The prototype operates on Raspberry Pi 4 with real-time inputs and cloud-based processing, validating system performance in a realistic usage environment.
13	Sustainable Development Goal (1-17) (Most prominent one goal)	SDG 10 – Reduced Inequalities • The project supports inclusive communication by enabling real-time sign language translation. • It improves accessibility for individuals with speech and hearing impairments, helping reduce communication barriers in education, healthcare, and public interaction environments.
14	Industry Practice followed for system development (Agile- SCRUM)	1. The project follows an Agile SCRUM-based development approach with sprint planning, execution, review, and retrospectives. 2. Incremental development, continuous testing, and regular feedback cycles are used to improve system reliability and performance.

Conference/Journal Publication/Patent Details (Mandatory, Target Publication a)

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Signature of Students: 1.
2.
3.

Signature of the Supervisor: